RFSAT Limited

STS

Shooting Target Scorer

User Guide

Version 1.54.0

For Android 8.0 and later

The current edition of this guide is always at
https://www.rfsat.com/download/STS-User-Guide.pdf

developed by Dr Artur Krukowski

Safety first

> **Range safety**
>
> Operate the app only from behind the firing line, with the firearm made safe and the range in a condition permitted by the range officer.
>
> Never handle a phone and a loaded firearm at the same time.

Nothing in this guide overrides the range officer. If scoring a string means walking forward, changing your position, or taking your attention off the firing point, do it when the range is cold and not before. The app is designed so that every step that needs your hands can wait.

If you are using live scoring, set the phone up on a stand before the relay begins and leave it alone. It is meant to watch the target so that you do not have to.

What STS does

STS scores shooting on paper and steel from the phone camera or an external video feed. You register the target face once, and every new hole is found, measured against the ring geometry of the selected competition target, and scored under the rules of the selected match.

The app reports the running total, the group centre and dispersion, and the sight correction — in clicks for the scope or red dot in your active profile set — needed to move the group onto the point of aim.

Disciplines covered

Air rifle and air pistol at 10 m, rimfire rifle and pistol at 25 m and 50 m, and centrefire rifle in .223, .308 and similar chamberings out to 1000 yards. Sixteen target faces and more than twenty rule sets ship with the app, and you can add your own.

Two ways to score

- From a photograph. Photograph the card after the string, load it, and score it. This is the more accurate route and the one to learn first.
- Live, as you shoot. The phone watches the target on a stand and adds each shot as it appears. Convenient, but it depends on a steady view and good light.

Either way, you can also choose WHO does the finding: the app's own algorithms, or an AI service — see “Scoring with an AI service”. The app's own algorithms are the default and the more precise of the two.

> **Before you rely on a score**
>
> STS is a training and practice aid. Official match results are determined by the appointed jury under the current rulebook of the governing body.
>
> Ring dimensions and rule parameters shipped with the app are nominal published figures and are editable — verify them against the rulebook in force for your competition.

Getting started

Work through this once. Everything here is remembered between sessions, so you should not have to do it again unless you change equipment.

Set your units and display

Open Settings. Under Display and units you can choose:

Setting	Options	Notes
Units	Metric (m, mm, cm) or Imperial (yd, in)	Metric is the default.
Theme	Dark, Day, Night — Green, Night — Red	Dark is default and saves power on OLED screens.
Full screen	On or off	Hides the system bars while shooting.
Diagnostic log	Show or hide	Adds a Log tab. Useful when something looks wrong.

The night themes exist to preserve dark adaptation. Under Night — Red the app keeps every element red, including the alignment guide, so nothing on screen costs you your night vision.

Every option on the Settings screen carries one line saying what to expect from it, with a More info link underneath for the reasoning and the measurements behind it. Nothing is hidden; it is simply not in the way of the switch.

Build a profile set

A profile set is your firearm, your load and your sight, kept together. The sight matters most: it is what turns a group offset in millimetres into a number of clicks.

1. In Settings, go to Profile sets and create a set.
2. Choose a Firearm from the catalogue, or add your own.
3. Choose a Load. This gives the app your bullet or pellet diameter, which is the scoring gauge.
4. Choose a Sight, and check its click value. If the app's correction ever moves your group the wrong way, use Invert elevation or Invert windage rather than doing the arithmetic in your head each time.

If your sight has no clicks

The catalogue carries two entries for sights that cannot be clicked: Built-in iron sight, and No sight. Choose one of these and the app stops offering turret clicks and tells you something you can act on instead.

- With a sight radius entered — the distance from the front sight to the rear sight — the correction is given as how far to move the rear sight, in millimetres. Measure it once with a ruler and it is then exact.
- Without one, the app says so and asks for the measurement rather than guessing. A sight radius runs from about 150 mm on a service pistol to 700 mm on an air rifle, and a plausible-looking default would produce a confident instruction to move the sight by the wrong amount.
- With No sight, nothing can be adjusted at all, so the group is reported as a hold-off: how far to aim off next time, and in which direction.

Note that a target pistol rear sight is an open sight AND click-adjustable. Choosing Built-in iron sight for a Morini or a Pardini would throw away clicks you actually have — pick the catalogue entry for the sight itself.

Choose a target face and a rule set

On the Home screen, set the Target, the Rules and the Distance for what you are about to shoot. The distance shown is the one this session is being scored at, which is not always the rule book's: a competition face shot at another distance in training is ordinary, and where the two differ the rule book's figure is shown in brackets beside it. The Home screen is a summary of your active setup: profile set, firearm, load, sight, target, rules and distance, all in one place.

> **Get the target face right**
>
> The face sets millimetres per pixel, the radius of a scoring area and which region counts as black. Choose a wrong one and score is wrong by roughly the amount the two faces differ — and if they differ enough, no hits are found at all.
>
> You rarely have to choose by hand. Both scoring screens can identify the face from the picture and adopt it for you.

The screens

Tab	What it is for
Home	Your active setup at a glance, and the place to resume a session in progress.
Session	Live scoring and camera capture. Also the route to scoring an uploaded photo.
Results	The shot plot, the score, the group statistics and the sight correction. Editing, export, and the second opinion.
Targets	The catalogue of target faces, with a preview showing the scoring rings and their numbers.
Rules	The catalogue of match rule sets, with every parameter shown.
Settings	Units, theme, profile sets, the detection options and the AI services.
Log	The diagnostic log, when you have switched it on.

Scoring a target from a photograph

This is the accurate route. Learn it first.

Taking the photograph

- Fill the frame with the card. The more of the picture the target occupies, the better every measurement is.
- Shoot square on. Small tilts are handled, but a steep angle is guesswork.
- Flat, even light. Avoid a hard shadow across the card and avoid direct flash, which blows out the paper and hides the holes.
- Do not brighten or adjust the photograph afterwards. The app reads colour as well as brightness, and an auto-levels pass throws away the difference it depends on.
- Photograph the clean card first, before the string, if you can — but see the note at the end of this section on what that is and is not good for.

Loading and registering

From the Session tab, choose Upload a target photo to score. The Import screen is laid out in four numbered steps that match what you have to do.

Choose the photographs

Pick the photo of the shot target. Optionally pick a photo of the clean target as well.

Course of fire

Confirm the Rules, the Target face and the Distance. These carry over from Home, so normally you only check them.

Register the target

Registration is the app working out where the target is in the picture and how big it is. Three buttons:

Button	When to use it
Identify target and register	Try this first, and it runs automatically when the picture loads. It fits the printed rings, identifies which catalogue face they belong to, adopts that face, and registers in one action.
Auto-detect the target	Use it when you have already chosen the right face and want it taken as given. It finds the black aiming mark, fits the printed rings for the selected face and draws them, exactly as the first button does — what it does not do is search the catalogue for a face that fits better.
Register	Manual. Drag the box onto the outer scoring ring yourself. Use it when the other two fail — a badly cropped or very low-contrast card.

If the card is at an angle, the Tilt and rotation controls let you correct it. Apply estimated tilt fills them in from what the app measured; Reset tilt clears them. Leave these alone unless the registration is visibly wrong. If the drawn circles line up with the printing across the card but not down it — or the other way round — that is not tilt but a stretched picture, and the Image aspect controls correct it: the app fills in the width and height percentages that would make the rings round, you press Apply and re-register, and the picture itself is stretched so the scoring geometry stays circular. Original picture puts it back.

Registration is always the app's own work, even when an AI service is scoring the card. Without knowing where the card is and how big it is there is no millimetre grid, and nothing could be drawn in the right place.

Score

Press Detect the hits and score them. What runs here depends on the Settings choice “On import, score with” — the app's own detection, or the AI service you have chosen. Choose whether to replace the shots already in the session or add to them. The shot distribution appears below, and Open the full results takes you to the Results tab.

Check the result before you trust it

On the Results screen, switch the plot to My photo. This lays the detected shots over your own photograph, and it is the only view in which a missed hole is obvious. On template a missing shot just looks like a shot you did not fire.

The photograph is kept with the session, so closing the app and coming back to it later leaves both the score and the picture where you left them. It is discarded only when you start a new session.

Add, move or delete shots as needed. Moving a shot keeps the app's sub-millimetre estimate of its centre where that estimate was good; deleting and re-tapping throws it away. Nudge rather than replace.

About the clean photograph

With two photographs the app can compare them and look for what changed, which in principle cancels everything printed and leaves only holes.

> **Measured, and it does not work hand-held**
>
> Two photographs of the same UNPUNCHED card, taken by hand a moment apart, differ by three to ten phantom holes. Registering each frame separately makes it worse, not better — so it is not camera shake.
>
> A printed ring line is about half a pixel wide in the flattened image, and no alignment good enough to make those lines cancel is available from a ring fit. Add a lighting gradient that shifts as the card is handled and the residue swamps a single new hole.
>
> It has not been tested from a FIXED MOUNT with the lighting undisturbed, which is what it was designed for. Until it has been, treat the clean photograph as unproven rather than as the accurate route.

Scoring live from the camera

Live scoring watches the target and adds each shot as it appears. It is convenient and it is less forgiving than a photograph — it depends on a steady mount, steady light and an unobstructed view.

Setting up

1. Mount the phone on a stand with a clear view of the target. It must not move for the rest of the string.
2. On the Session tab, choose the frame source: the phone camera, or a network stream such as an IP camera (an http://…/video or rtsp://… address). Pick MJPEG for an http address and RTSP for an rtsp one, then press Connect to the stream. The picture should appear in the viewfinder; capture the reference and start live detection from there as usual. The address is remembered, so it is typed once and is still there after a restart or an upgrade, and the app reconnects to it by itself when you come back to this tab. If no picture appears, the message says which stage was reached — nothing answered at the address, answered but no session, session but nothing decoded

— and the diagnostic log carries the whole handshake, including what the camera answered at each step. Two things fix almost every case: the phone must be joined to the camera's own Wi-Fi, and the app pins itself to that network because it has no internet and the phone would otherwise send everything over mobile data; and the address should be the one VLC shows under Tools > Codec information. A path is not required — the app tries the usual ones. If the address plays in VLC it will play here: both transports are supported, and the app falls back from UDP to TCP by itself. The stream view has to stay on screen for an RTSP source, because that is what the frames are decoded into — and if nothing arrives within ten seconds the app says why rather than showing a black rectangle.

3. Set the Analysis resolution. 1920 × 1080 is the default and the right choice for most cards. Higher costs speed; lower risks losing small holes.
4. Press Lock exposure and focus. If the camera re-focuses or re-meters mid-string, the picture changes underneath the detector and shots can be missed or invented.
5. Use the alignment guide to line the phone up — see below.
6. Press Capture clean target (reference), then Start live detection.

The alignment guide

The guide draws the rings of the selected face over the camera preview, at their true proportions. Line the printed rings up with the drawn ones.

Option	What it draws
Nothing	No overlay.
Simple crosshair	A cross at the centre of the frame.
Rings of the selected target	The rings of the face you have chosen, at true proportions.
Rings and crosshair	Both.

Guide size sets how much of the screen the rings occupy. A word in the corner tells you continuously whether the card in front of the camera matches the face you have selected:

Message	Meaning
Aim at the target	The rings cannot be fitted — cropped, too dark, or too steep an angle.
Checking...	A measurement is in progress.
Match OK	The card matches the selected face. The guide is drawn solid.
Wrong target face	The card does not match. The guide is drawn finely dashed, and the app will normally switch to the face it recognises and tell you it has.

The state is shown three ways at once — by colour, by whether the line is solid or dashed, and by the word — so it still reads correctly under the night themes, where the colour is deliberately left alone, and for a colour-blind shooter in daylight.

> **Do not force a match**
>
> The rings are drawn at the selected face's own proportions, so moving closer or further away resizes them all together. A different face can never be made to line up by changing distance. If the guide says the face is wrong, it is wrong.

During and after the string

A hole must appear in three consecutive frames before it is recorded. A flicker, a passing shadow or a moment of camera shake will not add a shot.

Score the target now (single frame) scores whatever is in view immediately, without waiting. Undo shot removes the last one recorded. When the string is over, go to Results and check the plot against the card.

Reading the Results screen

The plot

Shots are drawn as hollow circles with a small cross marking the centre of each, so that a shot on a ring line does not hide the line. Pinch to zoom and drag to pan; Reset view returns to the default framing. Zoom and pan survive editing, so you can nudge a shot without losing your place.

Control	What it does
My photo / Template	Switches between your own rectified photograph and the printed face. One or the other, never both. The photograph is kept with the session and is still there after the app has been closed and reopened.
Whole card / Scoring area	Whether the view fills with the scoring rings or the entire card.
Add	Tap the plot to place a shot the detector missed.
Move	Drag a shot to correct its position.
Delete	Remove a shot that was never fired.
Second opinion	Ask the AI service to check the card. Only when a key is set — see the next chapter.
Reset view	Return zoom and pan to the default.

Score, group and correction

- The headline is the total, and the maximum for the rule set where one applies. Inner tens are counted where the rules call for them.
- Group gives the centre of the group and its dispersion — how far your shots are from each other rather than from the middle.
- Sight correction converts the group offset into clicks for the sight in your active profile set, into a rear-sight movement in millimetres for an unclicked iron sight, or into a hold-off where there is no sight at all. When the group is already closer to the point of aim than one click would move it, the box says so and the line beneath it gives the residual — the two never disagree.
- Under that, the same correction as an angle: elevation and windage, in MRAD and MOA, in a column so the figures can be compared at a glance.
- Shot distribution shows how the shots fall across the rings.

Recording and sharing

Stage time is for practical disciplines where the score depends on it. Notes is free text — conditions, hold-off, anything worth remembering when you look back at this string in a month.

Share report produces a readable summary. Export CSV gives you the shot coordinates and scores for your own analysis. Finish and start a new session closes the current string and clears the plot for the next one.

Shots the app measured itself and shots placed by hand — or by an AI service — are recorded differently, and a shared report says which is which. A position that was not measured is never presented as though it had been.

Target faces and rules

Target faces

The Targets tab lists every face the app knows, with a preview and its full dimensions. The preview is drawn square so the face fills the width, with the scoring rings numbered as they are on the printed card — and the catalogue list stays visible underneath, which for the silhouette faces is the only way to tell an IPSC target from an IDPA one.

Sixteen faces ship with the app, including the ISSF 10 m air rifle and air pistol faces, the 25 m and 50 m pistol and rifle faces, the NRA and CMP high-power and smallbore faces, F-Class at 600 and 1000 yards, and the DSB/BDS faces.

Use this makes a face active. Copy & edit takes a catalogue face as a starting point for your own — this is how you add a club card or a face the app does not carry. Add my own target starts from nothing.

Rule sets

The Rules tab works the same way and shows every parameter of each match: shot count, scoring gauge, whether decimal scoring and inner tens apply, and the time limits where they matter. More than twenty rule sets ship with the app, from ISSF and NRA courses to NRL22, PRS, IPSC and IDPA stages.

> **Check the numbers against your rulebook**
> The dimensions and parameters shipped with the app are nominal published figures, not certified extracts from a governing body's tables. They are editable for exactly that reason. Before a competition matters to you, check the face and the rule set against the rulebook in force.

Scoring with an AI service

The app can send a photograph to an AI service and use what it reports. This is optional, off until you set it up, and it costs a fraction of a penny per card against your own account.

It exists because the app and a vision model are good at different things. The app MEASURES well — a hole it has found is placed to between 0.2 and 1.7 mm. It COUNTS badly: its usual failure is marking printing as shots. A vision model counts well and places badly: a position read off a picture carries several millimetres, which on a 10 m face can be a whole ring.

Choosing a service

Two are supported, and they are interchangeable. Both are sent the same flattened picture and the same question, and both are held to the same answer format, so a card scored by either arrives in the app in the same shape.

Service	Where its key comes from
Claude (Anthropic)	console.anthropic.com
OpenAI	platform.openai.com

Settings asks three separate questions about services, each saying what it governs. They are separate because they are separate decisions, and one shared setting could not express the useful combinations.

Setting	What it decides
On import, score with	What runs when you press Detect the hits and score them: Embedded, or a named service.
Second opinion asks	Which service the Second opinion button on Results calls.
Service to set up	Which service the key and model controls below are editing. It sends nothing on its own.

The first two are independent on purpose. Scoring with one service and checking with the other is the arrangement that makes a second opinion worth having; so is scoring with the app and checking with a service, which is the default.

Under Service to set up sits a model picker showing the models that service offers. The list is rebuilt when you change service, because an identifier from one means nothing to the other. Each list ends with Other, where you can type a model identifier by hand — a built-in list goes stale, and this way a newer model is never out of reach. Whatever you type must be able to read images and to answer against a schema.

Wherever the app is about to call a service, it names the one it is calling — “Asking OpenAI…”, “Scored by Claude (Anthropic)”. If a message names a service you did not choose, that is a bug worth reporting.

Setting it up

1. Get an API key from the service's own console and put credit on the account.
2. In Settings, under Service keys and models, set Service to set up to that service.
3. Press Set API key and paste it in. The line above says which service it will apply to.
4. Choose a model. The middle option of each list is the sensible default.
5. Then choose where it is used: On import, score with, and Second opinion asks, under AI assistance.

A key is kept for each service separately, so you can hold both and switch between them to compare. The screen lists every service and whether it has a key, so setting a second one is visibly not replacing the first.

> **The key is not your chat password**
>
> It comes from the service's developer console and bills that account. The password you sign in to the chat product with will not work here — they are different systems.
>
> Keys are stored encrypted on the phone and are never written to the diagnostic log. If the device cannot provide encrypted storage a key is NOT saved in plain text as a fallback; you are told instead.
>
> Paste a key as a single line. A key copied from a wrapped display carries a line break, which cannot go in a network request — the app strips whitespace and tells you when it has, but a clean paste is better.
>
> Keys survive an ordinary app upgrade. They are deliberately left out of device backup: the key that unlocks them stays in the phone's own keystore and does not travel, so a restored copy could never be read. After moving to a new phone, enter them again.
>
> Every use needs a working connection, which most ranges do not have.

Choosing what scores a card

Settings carries one line — On import, score with — and it decides what happens when you press Detect the hits and score them.

Choice	What happens
Embedded	The app's own detection runs, as it always has. A service is still available afterwards on the Results screen as a second opinion. This is the default and the more precise.
A named service	The app's detection does not run at all. That service finds and scores the shots. Registration is still the app's work, so the marks land exactly where you see them on your photograph. Falls back to Embedded if the service has no key — the picker still shows what you chose.

The second opinion

Under Embedded, the Second opinion button on Results asks the service to look at the same card and reports where the two disagree — how many it sees, how many the app marked, and which marks are in dispute.

You are then offered the action that fits the disagreement:

- When the app has marked MORE than the service sees, removal comes first. That is the common case, and the marks beyond the scoring rings are the likeliest to be wrong.
- When the service sees a shot the app missed, it is offered as a suggestion. The app then MEASURES that place before placing anything — the suggestion says where to look, and the measurement decides where the shot goes. If nothing hole-like is there, nothing is placed and you are told.

Neither is applied without your agreement unless you have asked for it. Settings has an option to let the AI answer override the app outright, and the import setting above hands it the whole job.

The second opinion uses the service named under Second opinion asks, whichever service scores your imports. The summary it shows names the service that answered.

> **What it cannot tell you**
>
> The service missing a real shot and the app inventing one look identical from the app's side. That is why removal is offered rather than performed, and why the dialog says so.
>
> Check the plot with My photo selected before agreeing to anything.

What to expect

On the app's own reference card — seven real shots, hand-scored at nineteen points — the service counts seven and is right. The app, with the option to look outside the scoring rings switched on, marks sixteen: the seven real shots and nine pieces of printing. Nothing in the app's own measurements separates those nine from the two genuine misses among them; counting is the only thing that has ever arbitrated it.

That is the honest case for the feature, and its limit. It will tell you reliably that a card has been over-marked, and where. It will not place a shot as precisely as the app does when the app can see it.

Detection options

These live in Settings under Detection algorithms. Each carries one line saying what to expect, with More info for the reasoning behind it. Each can be switched independently so its effect can be judged on your own targets. The defaults are sensible; you do not have to touch any of them.

Option	Default	What to expect
Find shots in the photograph itself	On	Leave on. Finds shots in the picture as it arrived, including inside the black aiming mark. With it off, centre hits are often missed entirely.
Estimate the background locally	On	Leave on. Copes with a card that is lit unevenly, which is every card photographed on a table. With it off, one end of the card reads as holes everywhere and the other as nothing.
Require a shot to have the profile of a puncture	Off	Fewer false hits: rejects printed marks that happen to look like shots.
Also find shots that missed the rings	Off	Shows shots that missed. They score nothing — but printing near the edge of the card may be marked too, and the app cannot tell those apart. Leave off unless you want to see where a flyer went.
Scale from circles fitted to each printed ring	Off	No visible difference on a flat card. May help on cards photographed at an angle.
Measure ring spacing only along the tilt axis	Off	Experimental. Helped on some test images and hurt on others. Leave off unless you are testing it.

Accuracy: what to expect, and what to check

This section is here because a scoring app that hides its limits is worse than one that has none. Read it once.

How the app is checked

Two kinds of card are used, and every release is measured against both.

A photographed ISSF 10 m Air Pistol card is built into the app and scored on every build. Its answer was established by hand, independently of the app's own code: the centre from circles fitted to the printed ring lines, the scale from those circles against the published ring radii. It carries five scoring shots — a 9, a 6, a 2 and two 1s, nineteen points — and two shots that missed the rings altogether.

The app scores that card nineteen, at both full and half resolution, with no false marks. Two of its shots are deliberately awkward and both have caught real faults: one sits inside the black aiming mark, and one straddles a ring line so that it scores a 6 only because the hole's edge breaks the line, where its centre alone would make it a 5.

Printed cards punched with a 5 mm drill at coordinates known to a tenth of a millimetre are used for everything the photographed card cannot test — shots at the edge of the black, shots off the card, and shots close enough together to merge. On the geometry card the app finds six of six and scores exactly right.

What is still imperfect

- Shots that merge. Two holes about a gauge apart are separated correctly. Three or more run together in a rosette are found as one region and then not reported at all, because splitting along one axis is meaningless for a round cluster. A tight ten-shot group in the bullseye is the case the app handles worst.
- Marks outside the scoring rings, when you have asked to see shots that missed. On the reference card that setting produces nine pieces of printing alongside the two genuine misses, and nothing the app measures separates them. With the setting off — the default — no mark can appear outside the rings at all.
- Distances from the centre read about a millimetre short at the outermost rings. On the test card this changed no score. On a shot sitting exactly on an outer ring line it could.
- Steeply angled photographs are the weakest case. The app corrects modest tilt well and guesses beyond that.
- Cards with no black aiming mark, and printed illustrations rather than real targets, are outside what the detection is designed for.
- The two-photograph difference method does not work on hand-held photographs — see the note in the photograph chapter.

What to do about it

Always look at the Results plot with My photo selected before you record a score. It takes a few seconds and it catches everything above. The app is built to be corrected: add what it missed, move what it placed badly, delete what was never there.

If whole card scores implausibly low, the usual cause is the wrong target face. Check face first, before anything else.

If the app has marked things that are not shots, the quickest remedy is to turn off Also find shots that missed the rings. If you need that setting on, the second opinion is the tool for pruning what it brings in.

Troubleshooting

Symptom	Likely cause and remedy
No hits found at all	Almost always the wrong target face. The face sets the size of the thing the app is looking for, so a face that is wrong by enough finds nothing rather than finding the wrong thing. Use Identify target and register.
The camera says “Wrong target face”	It is comparing the printed rings against the face you selected. Let it adopt the face it recognises, or change the face by hand. Do not try to make it fit by moving closer or further away.

Symptom	Likely cause and remedy
Marks appear where no shot was fired	Turn off Also find shots that missed the rings — that is where they come from. Turning on Require a shot to have the profile of a puncture helps as well. If you have an AI key, the second opinion will offer to remove them.
A shot in the bullseye is missed	Check that Find shots in the photograph itself is on. With it off, shots inside the black aiming mark are frequently missed.
Two shots close together are reported as one, or as none	Expected above about three merged holes. Add the missing shots by hand from the plot.
Some shots found, others missed	Check the light and the framing, then correct the plot by hand. Do not brighten or auto-adjust the photograph — that removes the colour difference the detector depends on.
The score changes between the two registration methods	Identify target and register is the more accurate. Use Auto-detect only when the first fails.
Live scoring misses shots	The camera almost certainly re-focused or re-metered. Press Lock exposure and focus before the string, and make sure the phone cannot move.
Sight correction moves the group the wrong way	Invert elevation or Invert windage in the sight profile.
The sight correction says no adjustment is needed	The group is closer to the point of aim than one click of your sight would move it. The line beneath gives the residual, which is information rather than an instruction.
My photo is greyed out	No photograph belongs to this session — it was scored by hand or from the camera, or a new session has been started since. Import a photo and it is kept from then on, including across a restart.
The app offers no clicks for an iron sight	Expected — enter the sight radius and it will tell you how far to move the rear sight instead.
“The reply was cut off”	The answer ran out of room. Try a smaller photograph or a different model.
“The API key was rejected”	It must be a key from the service's developer console, not the password you sign in to its chat product with. Check you set the key under the same Service to set up as the service actually being called.
“The key contains a space or a line break”	It was pasted from a wrapped display. Set it again as a single line.
The AI service is not asked at all	No key is set for the service named, or Second opinion is switched off in Settings. With a service named for import and no key, imports fall back to the app's own detection.
The wrong service appears to be used	Check both settings: On import, score with, and Second opinion asks. They are separate and can name different services. Every message names the service being called.
The keys have disappeared after changing phone	Expected. Keys are deliberately excluded from device backup, because the key that unlocks them stays in the old phone's keystore and a restored copy could never be read. Enter them again.
Something else looks wrong	Switch on the diagnostic log in Settings and repeat what you did. The log records what each stage measured and is usually enough to explain the result.

About

The current edition of this guide is published on the RFSAT portal at www.rfsat.com/download/STS-User-Guide.pdf. That address does not change between editions, so a copy of the link keeps working after the guide is reissued.

STS is developed by Dr Artur Krukowski, alongside VTB Vapor-Trail Ballistics. Both apps share a design principle: the physics and the rules are written down explicitly in the source, with their assumptions stated, so a shooter can see exactly why the app produced the number it did.

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STS is a training and practice aid. Official match results are determined by the appointed jury under the current rulebook of the governing body.

Ring dimensions and rule parameters shipped with the app are nominal published figures and are editable — verify them against the rulebook in force for your competition.

Operate the app only from behind the firing line, with the firearm made safe and the range in a condition permitted by the range officer.